

Reliable Measurement of Teachers' Instructional Practices with LLM Classifiers¹

Steffen Erickson and Vivian C. Wong

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Abstract

Large language models (LLMs) are increasingly used to score teachers' instructional practices from classroom transcripts. Most validation studies focus on whether LLMs accurately classify individual utterances, but this does not establish whether the aggregated teacher-level scores are reliable across lessons. We extend generalizability theory to LLM-based scoring systems to examine the reliability of teacher-level scores produced by fixed model–prompt configurations. Unlike human raters, a fixed LLM classifier applies the same scoring rule every time. Classifier performance can, however, vary systematically across teachers, creating errors that persist across their lessons. These systematic errors can appear as differences in teachers' instructional practice when they are classifier artifacts. Because these artifacts are reproducible, conventional reliability estimates may treat them as stable between-teacher differences and overstate how much reliable score variation reflects the intended construct. We develop construct-relevant reliability (CRR), a coefficient that estimates the proportion of observed teacher-score variance that is both reproducible across lessons and attributable to true differences in instructional practice. CRR can be projected to different lesson numbers and lengths from a small human-coded validation sample. Monte Carlo simulations examine how classifier accuracy, systematic scoring differences, and study design jointly affect CRR. We then use CRR to compare 240 model–prompt configurations for measuring open-ended questioning in elementary science classrooms. Across simulations and the empirical application, utterance-level accuracy alone is insufficient for selecting LLM-based scoring systems: configurations with similar classification performance differ in the reliability and validity of the teacher-level scores they produce.

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Introduction

Measuring teachers' classroom instruction is central to giving teachers feedback and to generating data on teaching effectiveness (Cohen & Goldhaber, 2016). Both uses depend on teacher-level scores that distinguish teachers consistently across conditions of measurement. A teacher's relative standing should not depend primarily on which lesson happened to be observed or which rater happened to score it. Traditional observation systems evaluate this reliability by observing teachers across multiple lessons and raters and using generalizability theory (G-theory) to distinguish stable variation among teachers from variation associated with conditions of measurement. These analyses can also be used to determine how many observations are needed per teacher to produce sufficiently reliable scores (Brennan, 2001; Cronbach et al., 1972; Hill et al., 2012; Kane & Staiger, 2012). Generalizability studies show that teachers typically must be scored across multiple lessons and raters, making traditional observation systems costly and difficult to implement with fidelity at scale. In response, researchers are increasingly turning to AI-based tools, including large language models (LLMs), to score instructional practices from audio, video, or classroom transcripts.

AI tools can reduce the burden associated with human raters, but teacher-level scores remain useful only if they consistently distinguish teachers from one another across lessons. Validation of machine learning-based instructional scoring systems commonly evaluates performance against human-coded labels at the utterance or observation-segment level (Martín-Hoz et al., 2025; Wang et al., 2025; Whitehill & LoCasale-Crouch, 2024; Xu et al., 2024). These analyses characterize model performance at the unit being classified, but they do not establish whether scores aggregated to the teacher level generalize across lessons. Few studies of

automated instructional scoring report teacher-level reliability; Hardy (2025) is a recent exception that applies generalizability theory to model-generated classroom ratings.

Replacing human raters with a fixed AI scoring rule introduces two forms of classifier error that matter for teacher-level reliability. In human-rated observation systems, differences among raters are one source of measurement error and can be reduced by averaging across multiple raters (Ho & Kane, 2013; Kane & Staiger, 2012). An AI-based system, by contrast, typically fixes a particular LLM and classification prompt and applies the same scoring function to all teachers' lessons. This makes the scoring rule highly reproducible, but it also changes how classification errors propagate across repeated observations. The first form is nonsystematic classification error that fluctuates across utterances and lessons, adding noise that can be reduced by averaging across more observations. The second is systematic variation in how the classifier functions across teachers or instructional contexts. We refer to this systematic variation as differential classifier functioning (DCF), by analogy with differential item functioning.

Systematic classifier error poses a particular problem for conventional reliability analysis because reproducibility can no longer be assumed to reflect the intended construct. Conventional reliability statistics treat reproducible between-teacher differences as signal. If a classifier systematically over- or under-scores particular teachers, those persistent classifier artifacts can therefore be incorporated into between-teacher variance, causing conventional reliability estimates to overstate the extent to which reproducible score differences reflect instructional practice. Unlike nonsystematic classification error, DCF does not diminish as additional lessons are observed, because the same teacher-specific classifier behavior recurs across observations. The distinction matters because the standard approach to increasing teacher-score reliability is to sample more observations per teacher. Additional observations can reduce nonsystematic

classification error, but they can also produce a more reliable score centered on the classifier's systematic bias toward a teacher rather than on the teacher's instructional practice.

Distinguishing these two forms of classifier error is therefore necessary for determining whether additional observations improve the measurement of instructional practice or merely make classifier-induced differences more reproducible. Large DCF can also make a classifier unsuitable for differentiating teachers' instructional practices even when the resulting teacher-level scores appear highly reliable.

In this paper, we extend G-theory to represent these sources of error from fixed LLM classifiers and introduce construct-relevant reliability (CRR), a reliability coefficient designed to distinguish reproducible variation in instructional practice from reproducible classifier artifacts. CRR estimates the proportion of observed teacher-score variance that is both reproducible across lessons and attributable to true differences in instructional practice. Whereas a conventional generalizability coefficient can treat stable classifier artifacts as reliable between-teacher variance, CRR restricts the numerator to variation associated with the intended construct. CRR can be estimated from a small human-coded validation sample and projected to different numbers and lengths of lessons, allowing researchers to evaluate a planned scoring system before applying it to a full corpus. The framework therefore links classifier performance at the utterance level to reliability at the teacher level and provides a common criterion for comparing alternative LLM classifier configurations used to generate teacher-level scores.

We use this framework to address three research questions:

1. How do classifier performance, differential classifier functioning across teachers and lessons, and lesson sampling jointly determine the reliability of teacher-level scores?

2. When can conventional reliability estimates overstate how reliably LLM-generated scores reflect true differences in instructional practice?
3. Can CRR be used to select model–prompt configurations that produce reliable teacher-level scores before deployment?

We address these questions in three steps. First, we derive the reliability coefficients and validate their behavior through Monte Carlo simulation. Second, we vary classifier performance, DCF, and sampling designs over plausible ranges to examine when classifier error affects teacher-level reliability and when conventional reliability estimates diverge from CRR. Third, we apply the framework to the problem that motivated this work: measuring the prevalence of open-ended questioning in elementary science classrooms. We treat 240 combinations of locally run open-weight LLMs and classification prompts as candidate scoring configurations, evaluate them on a human-coded validation sample, and use projected CRR to select configurations for application to the larger corpus. The empirical application demonstrates how reliability can become part of model and prompt selection rather than an evaluation conducted only after a scoring system has been deployed.

A Measurement Framework for Teacher-Level Prevalence Scores

The Measurement Procedure

We focus on instruments that tally discrete teaching moves and aggregate the counts over a lesson, for two reasons. First, such instruments are common, from classic protocols that code teacher–student interaction at fixed intervals (Flanders, 1970; Stallings, 1977) to frequency counts of explicit-instruction interactions (Gunn et al., 2021; Smolkowski & Gunn, 2012) and STEM protocols that aggregate discrete behaviors over a class (Eddy et al., 2015; Hora et al., 2013; Kranzfelder et al., 2019; Smith et al., 2013). Second, these are the quantities that

automated feedback systems compute and return to teachers. Research prototypes detect and count discursive moves in transcripts to generate feedback on classroom discussion (Jacobs et al., 2022; Suresh et al., 2022), and a commercial product, TeachFX, reports measures such as talk time, open-ended questions, and specific praise to districts nationwide (TeachFX, n.d.). A randomized trial with TeachFX found that such feedback increased teachers' use of focusing questions by about 20% (Demszky et al., 2025), so wider adoption is plausible.

Our running example classifies teacher utterances in fourth- and fifth-grade science lessons. It comes from scoring work for the Special Education Research Accelerator (SERA), which is studying a large corpus of classroom transcripts with low-inference measures of instructional and participation moves. We classify utterances with an utterance-level adaptation of the Science Discourse Instrument's *Ask* category, open-ended questions that generate productive science discourse (Fishman et al., 2017). Under hard scoring, lesson-level prevalence is the number of eligible teacher utterances classified as open-ended divided by the total number of eligible utterances. Under soft scoring, it is the mean positive-label score across those utterances.

We use off-the-shelf models: a pretrained generative LLM with a zero- or few-shot classification prompt, rather than an encoder trained to predict human ratings (e.g., Hardy, 2025). We use “prompt” in its broadest sense, everything the model receives besides the utterance to be scored: instructions, surrounding transcript context, worked examples, and the answer format. A prompt and a model together fully specify a scoring rule, so the instrument is fixed, auditable, and available to practitioners without labeled training data. Each utterance is embedded in the prompt, whose instructions constrain the answer to the labels “0” or “1,” and the soft score is the probability of label 1 given that the answer is one of the two labels, read from the model's next-

token logits in a single pass (Brown et al., 2020; Hendrycks et al., 2021; Sanh et al., 2022). This probability can be read as a confidence score, though its calibration depends on the model, prompt, and task and must be checked rather than assumed (Kadavath et al., 2022; Zhao et al., 2021).

Two properties of the procedure matter for measurement. First, with a pinned inference environment no tokens are sampled, so rescoring a transcript reproduces its scores; the instrument can be archived and audited. Determinism removes run-to-run noise but not classifier error. Among utterances that truly contain an open-ended question, some receive high scores and others low, and likewise among utterances that do not. Because a lesson's prevalence is the mean of these scores, this spread within each true class carries forward into lesson and teacher scores. It is this dispersion that our selection criteria treat as score noise. Second, because the output is a probability, thresholding it yields hard labels from the same run, so soft and hard scores can be compared directly. This system grounds the analysis, but nothing in the results depends on it: the coefficients below hold for any fixed classifier that emits utterance-level scores, whether an off-the-shelf LLM, a fine-tuned model, or a trained encoder.

From Utterance Labels to Teacher Scores

[Figure 1](#) traces the construction of a teacher score. The object of measurement is the teacher, whose practice is observed through a sample of lessons. Every eligible utterance in each lesson is passed through the same fixed classifier, which returns a probability (soft scoring) or a 0/1 label (hard scoring). The lesson score is the mean of the utterance scores; under hard scoring it is the fraction labeled positive, three of ten in the figure. The teacher score is the mean of the lesson scores.

The figure separates the two sources of variation the paper studies. Lessons are a *sampled* facet: instruction varies from lesson to lesson, so a teacher's lesson scores would differ even under perfect classification, and the teacher score estimates what the teacher would receive averaged over all the lessons they might teach. The classifier is a *fixed* condition: the same model and prompt label every utterance of every lesson, contributing no run-to-run variation. What it does contribute is the within-class dispersion described above and, potentially, systematic differences in how it treats particular teachers and lessons, which we will call differential classifier functioning.

The figure also fixes the distinction that organizes the paper. The accuracy question, asked at the utterance level, is whether each label matches a human coder; it is what classifier evaluation ordinarily answers. The reliability question, asked at the teacher level, is whether the scores would rank teachers the same way if a different sample of lessons had been observed. The two are related but not interchangeable. A classifier can be accurate on average yet yield unreliable teacher scores when lessons are few or short, and it can yield highly consistent teacher scores whose consistency reflects its stable treatment of particular teachers rather than their teaching.

Defining Teacher-Level Reliability

Generalizability theory defines reliability with respect to a universe of generalization: the set of conditions over which a score is meant to hold (Brennan, 2001). A teacher's universe score is the expected score over all admissible replications. In human-scored systems the replications vary by lesson and rater, and lesson-to-lesson variation, rater effects, and their interactions sometimes exceed persistent between-teacher variance (Boguslav & Cohen, 2023; Edwards & Soland, 2024; Hill et al., 2012; Ho & Kane, 2013; Kane & Staiger, 2012).

In LLM-based systems lessons remain a random facet; a recent variance decomposition of LLM-generated observation scores found that lesson-to-lesson variation within teachers persists when the rater is a model (Hardy, 2025). The scoring configuration could be treated as random, asking how scores generalize across a universe of admissible configurations, or as fixed. We treat it as fixed for three reasons. There is no substantively meaningful universe of admissible configurations to sample from: the set of models, prompts, and versions is large, changing, and full of configurations that would fail basic standards. Fixed treatment matches deployment, in which one prespecified configuration scores every teacher and the inference concerns new lessons scored by that rule; vetting candidate model–prompt combinations against human scores before committing to one is an emerging practice (Rubin & Grissom, 2026). And variation across vetted configurations is informative at the selection stage, where large differences among otherwise plausible configurations signal a need for further validation, rather than noise to fold into an operational error term.

The fixed treatment narrows what the reliability coefficient means. Classifier errors that fluctuate across lessons enter the error variance and lower the coefficient. Classifier biases that are stable within a teacher enter the teacher’s universe score, where they can leave the coefficient unchanged or raise it. Scores must therefore be read as prevalence *as measured by the selected system*: the coefficient describes the consistency of a teacher’s scores across lesson samples under that configuration, not evidence that persistent differences reflect teaching rather than classifier artifact. We pair reliability estimates with diagnostics for differential classifier functioning, and the coefficients below make the classifier’s role explicit.

Reliability Coefficients for a Fixed Classifier

The score model. Let π_{pl} be the true prevalence of the practice in lesson l of teacher p , and let π_p , the average over the teacher's universe of lessons, be the object of measurement. Because every utterance in a lesson is scored, utterances are a census of that lesson and need not be a separate facet. The classifier's scores behave as draws from two class-conditional distributions, one for utterances that contain the practice and one for those that do not (Green & Swets, 1966; Pepe, 2003). Write μ_1 and μ_0 for the mean score in each class. We present the coefficients needed to define CRR and to read the simulations and the empirical example; the full set, including the lesson-level and covariance-inclusive forms, is derived in the technical appendix. The expected lesson score is then a linear function of true prevalence,

$$q_{pl} = A_{pl} + J_{pl} \pi_{pl},$$

where $A_{pl} = \mu_0$ is the negative-class intercept, the mean score given to utterances that are not open-ended questions, and $J_{pl} = \mu_1 - \mu_0$ is the class separation. Under hard scoring A is the false-positive rate and J is Youden's index, sensitivity plus specificity minus one, so every result below covers hard classification as a special case. The observed lesson score is $\hat{q}_{pl} = q_{pl} + \varepsilon_{pl}$, where the realization error ε_{pl} arises because a finite transcript's average score fluctuates around its expectation. Its variance is proportional to $1/n_{pl}$, the number of utterances, and depends on the within-class score variances and on J . It is largest under hard scoring, because a bounded score's variance is maximized when the score takes only the values 0 and 1, so aggregating soft scores lowers the noise floor at equal separation. The teacher score averages n_L observed lesson scores, and its expectation over the teacher's universe of lessons, τ_p , is the universe score whose reproducibility reliability measures.

The reliability coefficient. The generalizability coefficient for teacher scores is

$$E\rho^2 = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma_{L:P}^2/n_L},$$

where σ_τ^2 is the variance of teachers' universe scores and $\sigma_{L:P}^2$ is the lesson-within-teacher variance of observed lesson scores. The latter has two parts: systematic lesson-to-lesson variation in the expected score, and the realization noise, which declines with lesson length. Reliability therefore rises with the number of lessons and, through the noise term, with their length.

Constant classifier functioning. Suppose first that A and J are the same for every teacher and lesson. Then each teacher's universe score is $\bar{A} + \bar{J}\pi_p$, and

$$E\rho^2 = \frac{\sigma_\pi^2}{\sigma_\pi^2 + \frac{\sigma_{\pi,L:P}^2 + \bar{\sigma}_\varepsilon^2/\bar{J}^2}{n_L}},$$

where σ_π^2 and $\sigma_{\pi,L:P}^2$ are the between-teacher and within-teacher variances of true prevalence and $\bar{\sigma}_\varepsilon^2$ is the average realization variance. Two things follow. If lessons were infinitely long the noise term would vanish and classifier accuracy would not affect reliability at all: a constant intercept shifts every score equally and a constant separation rescales all variances equally, and reliability, a squared correlation, is invariant to origin and unit (Lord & Novick, 1968). With lessons of finite length, accuracy matters through the ratio $\bar{\sigma}_\varepsilon^2/\bar{J}^2$: weak separation compresses the signal while leaving the noise floor unchanged, which is equivalent to inflating the noise by $1/\bar{J}^2$. A classifier with $\mu_1 = .70$ and $\mu_0 = .20$ has $\bar{J} = .50$ and quadruples the noise contribution. Accuracy matters most when lessons are short and separation is small. It also matters for scale: a small $\bar{J}$ compresses observed differences and makes the score a poor estimate of prevalence even when teachers' ordering is reliable.

Differential classifier functioning. The constant case assumes the instrument functions identically regardless of whose teaching it scores. We call systematic variation in A_{pl} or J_{pl}

across teachers or lessons *differential classifier functioning*, by analogy with differential item functioning, in which examinees at the same trait level respond differently to an item (Holland & Wainer, 1993; Millsap, 2011), and with differential rater functioning, in which a rater's severity varies across ratees (Myford & Wolfe, 2003). Variation in the intercept shifts the classifier's baseline, like uniform DIF; variation in separation changes the relationship between true class and score, like nonuniform DIF (Swaminathan & Rogers, 1990). Epidemiological work on misclassification shows that differential misclassification of a binary variable can bias derived quantities in either direction (Bross, 1954; Jurek et al., 2005). DCF is also plausible here. Machine-learning audits document error rates that differ across groups (Buolamwini & Gebru, 2018), and automated speech recognition, which produces the transcripts being scored, has higher word error rates for Black speakers than for white speakers (Koenecke et al., 2020). Stable speaker-associated transcription error is a concrete mechanism for teacher-level DCF; recording quality, topic, and discourse structure are mechanisms for lesson-level DCF. Whether DCF is present in a given system is an empirical question, which is what the diagnostics below are for.

Decompose the classifier parameters into a fixed part, a stable teacher-specific part, and a lesson-specific part: $A_{pl} = \bar{A} + a_p + a_{pl}$ and $J_{pl} = \bar{J} + j_p + j_{pl}$, with variances $\sigma_{a,P}^2, \sigma_{j,P}^2$ at the teacher level and $\sigma_{a,L:P}^2, \sigma_{j,L:P}^2$ at the lesson level. The two levels act differently. Lesson-level DCF averages out of the universe score and adds only to error, so it lowers reliability just as ordinary lesson-to-lesson variation does, and more lessons reduce it. Teacher-level DCF does not average out. It enters the universe score, so the universe-score variance becomes

$$\sigma_\tau^2 = \bar{J}^2 \sigma_\pi^2 + \sigma_{a,P}^2 + \sigma_{j,P}^2 (\mu_\pi^2 + \sigma_\pi^2),$$

where μ_π is the mean prevalence. Only the first term reflects teaching. The second is reproducible variation in false positives; the third is reproducible variation in separation, amplified by the level of the practice. Because all three are reproducible across lessons, the coefficient counts all three as true-score variance and *rises* with teacher-level DCF. Teachers can also be reordered, not merely rescaled, because a teacher-specific slope is not a monotone transformation of prevalence. The share of universe-score variance that reflects prevalence,

$$\lambda = \frac{\bar{J}^2 \sigma_\pi^2}{\sigma_\tau^2},$$

separates construct-relevant from construct-irrelevant reliable variance: the coefficient overstates construct-relevant precision whenever $\lambda < 1$. The two DCF terms behave differently with base rate. Intercept DCF adds a constant regardless of how common the practice is, whereas slope DCF scales with $\mu_\pi^2 + \sigma_\pi^2$, so for rare practices false-positive stability is the binding requirement and for common ones slope stability is. The generalizability coefficient with DCF at both levels keeps the form of $E\rho^2$ above, with σ_τ^2 replaced by the expression just given and the lesson-level DCF variances added to the error term; it appears as the denominator of the CRR expression in the next section, and the technical appendix gives the general form.

Covariance among classifier parameters and prevalence. The expressions above treat the intercept deviation, the separation deviation, and prevalence as uncorrelated. Nothing requires this, and each possible covariance has a natural reading. The classifier's false-positive rate may track how often a teacher actually uses the practice (a questioning style that pervades a teacher's talk). Its separation may depend on prevalence; a negative covariance describes *saturation*, a classifier whose positive and negative scores draw together for high-prevalence teachers, so that it distinguishes least among the teachers with the most to be distinguished. And the intercept and separation can covary with each other: because scores are bounded at 1, a

teacher for whom the classifier's negative-class mean runs high leaves less room for the positive-class mean above it, so a higher intercept tends to come with a lower separation. Admitting these covariances adds signed cross terms to σ_t^2 and to the error variance (technical appendix). The coefficient keeps its form, but λ needs a more general definition, because part of the DCF variance is now aligned with prevalence. We therefore define λ as the squared correlation between a teacher's universe score and true prevalence, which is what the zero-covariance expression already measured. Write $\sigma_{a\pi,P}$ and $\sigma_{j\pi,P}$ for the teacher-level covariances of the intercept and the separation with prevalence. The slope of the universe score on true prevalence is then

$$\beta = \bar{J} + \frac{\sigma_{a\pi,P} + \mu_\pi \sigma_{j\pi,P}}{\sigma_\pi^2},$$

the nominal separation plus a term the covariances supply, and $\lambda = \beta^2 \sigma_\pi^2 / \sigma_t^2$. It lies between 0 and 1 without truncation, equals the expression above when the covariances are zero, falls under saturation ($\sigma_{j\pi,P} < 0$), and rises when the classifier's artifacts happen to align with prevalence. Alignment improves teachers' ordering in the population where it was measured, but it is a property of that population rather than of the instrument, and a reliability figure that depends on it holds only where the alignment holds. Apply the same classifier to teachers whose prevalence and classifier errors are not aligned in the same way, and the reliability falls back toward what the classifier's own properties support; the reported figure would have overstated it.

Selecting a Reliable Classifier

A user needs a single quantity that rewards the two properties reliability depends on, reproducibility across lessons and fidelity to true prevalence, evaluated at the design the user intends to run. In generalizability theory a G study estimates the variance components of a measurement procedure, and a D study uses them to project reliability under a planned design.

CRR is a D-study coefficient of that kind: it is projected, not observed, and it is specific to the planned number and length of lessons. We define the *projected construct-relevant reliability* as the product $\text{CRR} = \lambda \times E\rho^2$ at a planned number of lessons n_L and lesson length n . Under zero covariance the product simplifies to

$$\text{CRR}_{zc} = \frac{\bar{J}^2 \sigma_\pi^2}{\bar{J}^2 \sigma_\pi^2 + \sigma_{a,P}^2 + \sigma_{j,P}^2 (\mu_\pi^2 + \sigma_\pi^2) + \frac{(\bar{J}^2 + \sigma_{j,P}^2) \sigma_{\pi,L:P}^2 + \bar{\sigma}_\varepsilon^2}{n_L}}.$$

The numerator keeps only prevalence-driven variance; the denominator adds the two ways a configuration can fail, reproducible classifier artifact and lesson-sampling error at the planned number of lessons. CRR is magnitude-aware: a classifier with no separation has no numerator, so a clean decomposition cannot carry a weak classifier past any floor, and a strongly separating classifier cannot pass if teacher-level DCF or the noise floor consumes its error budget. A configuration is selected when CRR exceeds a floor chosen for the use at hand.

With covariances admitted, CRR becomes the projected squared correlation between the n_L -lesson teacher score and true prevalence, the quantity selection requires. The difference from the zero-covariance version is captured by one signed number, the *alignment diagnostic* $\delta_\beta = \beta - \bar{J}$: the part of the slope that the covariances supply, in the same units as $\bar{J}$. When $\delta_\beta = 0$ the two definitions coincide and CRR_{zc} is exact. When $\delta_\beta < 0$ (saturation) CRR_{zc} is optimistic. When $\delta_\beta > 0$ the covariance-inclusive CRR is higher, but the advantage reflects population-specific alignment and does not hold in other populations. Among candidates with similar CRR, the one with δ_β nearest zero is the better instrument: its reported reliability means what it says in any population, the analog for a classifier of measurement invariance.

Two estimators follow; we write CRR for the covariance-inclusive coefficient and CRR_{zc} for its zero-covariance form. CRR_{zc} is stable and exact when $\delta_\beta = 0$ but rests on an assumption

the data can contradict, overstating construct-relevant reliability under saturation and understating it under positive alignment. The covariance-inclusive estimator of CRR estimates the right quantity, but it has two weaknesses of its own. The first is statistical. Because CRR is a squared correlation, its numerator is a squared covariance estimate, and squaring adds the estimate's own sampling variance, so it is biased upward in principle; the bias is negligible when the true covariance is large relative to its sampling noise and matters mainly when the true CRR is near zero or teachers are few, and with few teachers the estimate is also imprecise. The second is a pathology of the estimand itself. Because β is $\bar{J}$ plus a covariance term, a classifier with almost no separation can still report a high CRR if its false-positive rate happens to rise with teachers' true prevalence: the covariance term supplies the slope that separation does not, the DCF cross terms cancel most of the universe-score variance that is not aligned with prevalence, and a squared correlation cannot tell a classifier that measures the practice from one whose errors mimic it. Such a classifier separates nothing and would fail CRR_{zc} , which is why separation must be checked alongside the covariance-inclusive ratio.

We therefore recommend evaluating a classifier on five quantities together. (1) CRR_{zc} is the ranking statistic. (2) δ_β is the assumption check for each candidate. (3) CRR is the covariance-inclusive estimate of the correct estimand, reported alongside the ranking statistic. (4) $\bar{J}$ detects the failure case just described, in which CRR is high without separation. (5) $\bar{A}$ matters when the score is read as a level rather than a ranking: a high intercept biases estimated prevalence upward even though it cancels from every reliability ratio.

The procedure follows from the panel. Rank candidates on CRR_{zc} and keep those above the floor. Require a minimum $\bar{J}$, so that a candidate's ratio reflects separation between the classes rather than covariance alone. Then inspect δ_β for the survivors, require that CRR clear the floor

as well, and among near-ties prefer the candidate whose δ_β is nearest zero, because its reported reliability then depends least on the sample it was estimated in.

Two qualifications apply, both consequences of estimating variance components from a small number of teachers; the simulation studies below establish their size. First, the components are noisy, and negative estimates are truncated at zero. When both teacher-level DCF components truncate, $\lambda = 1$ by construction and CRR_{zc} becomes an upper bound rather than an estimate; the panel flags this case. Second, δ_β has little power at small samples, so a single value should be read loosely, whereas a consistent sign across candidates is informative. With a dozen teachers the panel is a screening device; with fifty or more it approaches an inferential one. The sampling behavior of all three estimators, CRR_{zc} , CRR , and $\hat{\delta}_\beta$, at both sample sizes is examined in the simulation studies.

Estimating the Components

All of the quantities the coefficients require, the means $\bar{A}$, $\bar{J}$, and μ_π , the teacher-level and lesson-level variance and covariance components, and the realization variance, are estimable from a validation sample in which each of N_p teachers contributes K hand-coded lessons. The estimation is a multivariate generalizability study on the lessons-within-teachers design with three outcomes per lesson (Brennan, 2001), and it proceeds in three steps.

First, within each coded lesson, the gold labels split the utterances into negatives and positives. The mean classifier score in each class gives $\hat{A}_{pl}$ and $\hat{\mu}_{1,pl}$, hence $\hat{J}_{pl} = \hat{\mu}_{1,pl} - \hat{A}_{pl}$, and the share of positives gives $\hat{\pi}_{pl}$. Each lesson thus contributes one observation of the triple $(\hat{A}_{pl}, \hat{J}_{pl}, \hat{\pi}_{pl})$, and the means of the three over all lessons estimate $\bar{A}$, $\bar{J}$, and μ_π . The same utterances supply the pooled within-class score variances, from which the realization variance at any planned lesson length follows in closed form.

Second, the triples are decomposed into a teacher-level covariance matrix and a lesson-level covariance matrix by the multivariate analog of the one-facet nested G study: the between-teacher matrix of teacher-mean triples and the within-teacher matrix of lesson deviations are equated to their expectations and solved for the two component matrices, exactly when teachers contribute unequal numbers of lessons. Variances and covariances alike are the output; the zero-covariance coefficients read only the diagonals. Because each lesson's triple is estimated from a finite number of coded utterances, the within-teacher matrix mixes genuine lesson-to-lesson variation with utterance sampling error. The sampling part has a known structure (technical appendix) and is subtracted so that the lesson-level components are not inflated; the teacher-level matrix needs no correction because the same term cancels in the between-minus-within difference. The realization variance is not a component of this decomposition. It depends on lesson length, so it is computed from the pooled within-class score variances at the planned lesson length and added in the D study, where the lessons to be scored may be longer or shorter than the lessons that were coded.

Third, the coefficients follow by substituting the estimated components into the closed forms at the planned number and length of lessons. Sampling uncertainty for the whole panel comes from resampling teachers with replacement and carrying each replicate through the same substitution. The `mvgstudy` package, written by the first author for multivariate G and D studies in Stata, implements the estimation and the projection, and its `mvcorr` command reports CRR and the companion panel (technical appendix).

A validation sample without lesson replication, in which each teacher contributes a single coded utterance set, as in the empirical example below, can still supply the teacher-level components: the across-teacher covariance matrix of the teacher-level triples is disattenuated for

within-teacher sampling error in the same way. What it cannot supply are the lesson-level components, which must then enter as design assumptions about within-teacher prevalence variation and lesson-level classifier functioning.

Simulation Studies

The simulations serve three purposes. They validate the closed-form coefficients of the previous section by showing that Monte Carlo reliability matches them when their assumptions hold (RQ1); they quantify how reliability responds to the factors a researcher controls or samples over (RQ1, RQ2); and they characterize when the coefficient overstates construct-relevant reliability (RQ2). They relax the framework’s assumptions one at a time: constant classifier functioning first, then differential functioning, then zero covariance between functioning and prevalence. Data are generated at the utterance level, so that realization error and its dependence on lesson length emerge from the process rather than being injected (Lord, 1955; Brennan & Kane, 1977). Each teacher receives a latent propensity to use the practice, each lesson’s prevalence deviates around it, lesson lengths follow a negative binomial calibrated to the corpus used in the empirical example, and each utterance’s true label is a Bernoulli draw at the lesson’s prevalence. The simulated classifier emits a beta-distributed soft score for every utterance with class means μ_1 and μ_0 and a concentration parameter that governs within-class spread (Kull et al., 2017): low concentration yields *diffuse* scores spread over the unit interval and high concentration yields *confident* scores clustered near the class means. We vary concentration because LLM classifiers differ in how confidently they score: at the same operating point, one model’s probabilities cluster near 0 and 1 while another’s spread across the interval. Scores are aggregated exactly as in the framework, and a thresholded twin of every score is computed at the

Youden-optimal cutpoint (Youden, 1950), so each cell yields a paired soft-versus-hard comparison.

The reference condition is calibrated to the corpus used in the empirical example: a classifier at the operating point of the configuration with the highest Youden index in the validation scoring (sensitivity .855, specificity .743), 51 teachers, twelve lessons per teacher, a mean of 81 utterances per lesson, and a mean prevalence of .10, with design factors swept one or two at a time around it. Conditions are labeled by a target sensitivity–specificity pair, because sensitivity and specificity are the standard summary of a binary classifier’s performance (Pepe, 2003), and the class means are solved so that thresholding at the Youden cutpoint achieves the target exactly. This makes the diffuse and confident models comparable at equal *reported* accuracy: a given overlap between the class distributions can come from widely spread scores with distant means or from tightly clustered scores with close means, and the two models realize the same operating point the two ways (separation $J = .54$ diffuse versus .23 confident at the reference).² Each cell runs 1,000 Monte Carlo replicates of the 51-teacher validation sample, from which the coefficient is estimated as a practitioner would; an auxiliary 10,000-teacher run that computes replication reliability (what the coefficient claims to estimate) and construct-relevant reliability (the squared correlation with true prevalence, what the researcher wants); and the closed forms evaluated at the cell’s true parameter values.

Constant classifier functioning. The constant-functioning simulation (304 cells) holds the class means constant across teachers and lessons. The closed form tracks the replication benchmark with mean error $-.0004$ and maximum absolute error .023, and the estimator carries only the expected small-sample bias (Figure 2). The formulas can therefore stand in for

² The two models have identical confusion matrices and different measurement properties; accuracy statistics cannot distinguish them, and the reliability coefficients do.

simulation when planning a measurement design: a user who wants to know how reliable teacher scores will be under a given number and length of lessons, for a classifier with given separation and intercept, can evaluate the coefficient directly rather than simulate it. Class separation is the dominant classifier characteristic at short lessons and fades as lessons lengthen, as the constant-functioning result predicts, because separation enters only through the noise term (Figure 2): reliability climbs steeply in J when lessons average 25 utterances and the curves flatten by 400. The intercept behaves as a pure noise effect (Figure 3): at equal separation, raising A from .05 to .35 costs .12 in reliability under diffuse scoring and .04 under confident scoring. At the calibrated reference, twelve lessons of 81 utterances yield reliability of .72 to .74, and reliability falls quickly with fewer lessons (.28 at two, .45 at four, .63 at eight; Figure 4, left). The four-lesson value matters because a handful of lessons per teacher is a typical observation budget; at that budget even a well-behaved classifier delivers about .45 at typical lesson lengths, so a reliability floor in the .6 to .7 range requires longer lessons or more of them. Raising mean prevalence from .10 to .40 raises reference reliability from .72 to .86 (Figure 4, right): rare practices are intrinsically harder to measure. Neither aggregation rule dominates (Figure 5). Soft aggregation beats matched-operating-point thresholding by .05 to .08 at the reference, but under confident scoring at asymmetric operating points the class means nearly coincide and thresholding wins by .20 or more; the closed forms, fed a candidate's actual class-conditional moments, resolve the choice case by case.

Differential classifier functioning. The DCF simulation (284 cells) lets the class means vary across teachers and lessons with random effects on the probability scale at three magnitudes; at the middle magnitude the SDs are about .035 in μ_0 and .065 in μ_1 . Teacher-level DCF produces the central result (Figure 6). At the middle magnitude with twelve lessons, the

generalizability coefficient a practitioner would estimate from the LLM-generated scores averages .85, and the scores' true replication reliability is .86, so the estimate is accurate for what it claims to measure. But the squared correlation of those scores with true prevalence is only .33, and $\lambda = .34$: two-thirds of the reproducible variance the coefficient certifies is classifier artifact. No statistic computed from the LLM scores alone reveals this. The coefficient is high, it agrees with replication, and it rises with more lessons, exactly as a sound instrument's would. The reason is that reproducibility, which the coefficient measures, is evidence of validity only when errors are not reproducible; a fixed classifier repeats its teacher-specific errors in every lesson, so the coefficient can no longer distinguish the reproducible variance that is teaching from the reproducible variance that is the classifier. Sampling more lessons makes it worse (Figure 7): as lessons go from 2 to 16 the coefficient climbs from .48 to .89 while construct-relevant reliability plateaus at .34, so more lessons raise the coefficient without raising construct-relevant reliability. Lesson-level DCF, by contrast, behaves as ordinary error; the coefficient and both benchmarks fall together, and more lessons reduce it. Which parameter drifts matters as much as where it drifts (Figure 9). When the intercept alone varies across teachers, at the base rate of .10, λ falls to .35 and the rank correlation between scores and true prevalence is only .55, so teachers are reordered; when the separation alone varies, λ stays at .93. The asymmetry follows from the analytic result that slope DCF is scaled by $\mu_{\pi}^2 + \sigma_{\pi}^2$, which is small for a rare practice. For rare practices, then, stability of the false-positive rate across teachers is the binding requirement on a classifier. The closed forms extended for DCF, the coefficients of the differential-functioning subsection, track the benchmarks with mean error +.002 (Figure 8). The largest deviations, up to .07, occur in cells where the classifier parameters and prevalence are correlated, violating the

zero-covariance assumption; those are the cells the covariance simulation addresses, and the violation is visible in the same covariance estimates the panel reports as δ_β .

Covariance between classifier functioning and prevalence. The covariance simulation (57 cells) drops the zero-covariance assumption and lets the classifier's intercept and separation covary with true prevalence at the teacher and lesson levels, at a reference cell calibrated to the empirical example ($\bar{A} = .15$, $\bar{J} = .55$, $\mu_\pi = .20$, $\sigma_\pi = .07$; four lessons of 150 utterances; $\text{CRR}_{zc} = .55$). The truth is the realized squared correlation between the four-lesson teacher score and true prevalence, from a 10,000-teacher benchmark; the covariance-inclusive closed forms match it with a mean gap of .006 (maximum .026; [Figure 10](#)). [Figure 10](#) shows how the three coefficients diverge as the covariance grows. As the alignment diagnostic δ_β moves from $-.42$ to $+.42$, the truth runs from .09 to .85 while CRR_{zc} stays at .55, because the zero-covariance formulas omit the covariance. CRR_{zc} is exact only at $\delta_\beta = 0$. A user must decide how much covariance to tolerate before rejecting a configuration. δ_β expresses that tolerance in units of the separation: because it is the gap between the slope of the teacher score on true prevalence and the nominal separation $\bar{J}$, a tolerance of $\pm .20$, chosen below to equal one sampling SD of the estimate, accepts slopes between .35 and .75 when $\bar{J} = .55$. Inside that band CRR_{zc} overstates the truth by at most .16; at $\delta_\beta = -.42$ it overstates by .46. The $\delta_\beta = -.42$ cell is severe saturation: the classifier's separation shrinks and its false-positive rate falls for high-prevalence teachers, so the score rises with true prevalence at only about a quarter of the nominal separation $\bar{J}$. The tolerance is two-sided because the zero-covariance coefficient is inexact in either direction: negative values of δ_β make it overstate the truth and positive values make it understate the truth, though understatement is the conservative error. The overstatement depends on δ_β alone, not on which parameter carries the covariance. The replication coefficient $E\rho^2$ overstates the truth everywhere

and rises on both sides of zero, since stable covariance of any sign looks like teacher variance to a G study.

Lesson-level covariance, which δ_β does not detect, moves the truth by at most $\pm.05$ at four lessons and shrinks with more lessons, unlike the teacher-level effect (Figure 11).

At twelve teachers CRR tracks the truth closely on average with a sampling SD of about .15, and $\hat{\delta}_\beta$ is unbiased with SD .15 to .21, so the $\pm.20$ tolerance is about one SD (Figure 12). The pathology cell described earlier, a classifier with almost no separation ($\bar{J} = .05$) whose intercept tracks prevalence, reports $\text{CRR} = .62$ against $\text{CRR}_{zc} = .004$, confirming that $\bar{J}$ must accompany the covariance-inclusive coefficient.

Correlated scores within lessons. The coefficients assume utterance scores are conditionally independent within a lesson. A further simulation relaxes this by letting lessons differ in discourse composition, so that the scores of a lesson's utterances are correlated. Under that dependence the independence formula overstates reliability by up to .18. The overstatement does not require a new coefficient, because dependence within a lesson has the same effect as lesson-level DCF: it makes the classifier's effective intercept and separation vary from lesson to lesson, and the coefficient that includes lesson-level DCF recovers the benchmark (technical appendix). Within-lesson dependence is therefore absorbed by the framework as already stated, provided the lesson-level components are estimated or assumed.

Summary of the Simulation Results

Together the simulations answer RQ1 and RQ2. Reliability is governed by lessons per teacher (the design factor with the largest effect), class separation relative to the noise floor (binding at short lessons), the intercept (a pure noise penalty), lesson-length heterogeneity, and prevalence, and the closed forms show how these trade off. Among classifier characteristics,

stability of functioning across teachers dominates everything else, because it is the one failure that reliability coefficients not only miss but reward; and when functioning also covaries with prevalence, the zero-covariance CRR departs from the truth in proportion to δ_β while the replication coefficient overstates in both directions. The implications for selecting a classifier follow directly. Utterance-level accuracy and the replication coefficient are both insufficient, because each rewards a classifier whose errors are stable across a teacher’s lessons. Selection therefore requires a human-coded validation sample with enough teachers to estimate the DCF components, and a criterion that separates construct-relevant variance from reproducible artifact. Our proposed method for assessing candidate LLM classifiers does this in stages: CRR_{zc} ranks configurations for the planned design, the separation $\bar{J}$ and the alignment diagnostic δ_β check whether that ranking can be trusted, and the covariance-inclusive CRR confirms that the survivors clear the floor. Among configurations with similar CRR_{zc} , the one with δ_β closest to zero is the better instrument, because its reported reliability does not depend on the prevalence distribution of the validation sample. Finally, lessons per teacher determines whether any configuration clears the floor, so the criterion certifies a configuration for a specified deployment rather than absolutely. The empirical example applies this procedure.

Empirical Example

The empirical example applies the framework to the task that motivated it: selecting an LLM configuration to measure the prevalence of open-ended questions in elementary science instruction. Its purpose is to demonstrate the workflow and the method, not to establish that the models we assessed are suitable for the task. The validation sample is small (twelve teachers’ hand-coded segments), so every per-configuration estimate is imprecise (sampling SD about .15 for CRR at twelve teachers in the covariance simulation), and the selection it supports is a

screening of candidates rather than a confirmatory verdict. The sample also has no lesson replication, so the lesson-level error components enter as assumptions rather than estimates. The human codes that serve as the reference standard are themselves a measurement with its own error, which the framework does not model; we treat them as the truth for the purpose of the demonstration, so that “true prevalence” in the example means prevalence as the human coders would have scored it.

Data

Transcripts come from SERA, which collected the recordings as part of its study of science instruction for elementary students with learning disabilities (Therrien et al., 2025) and coordinated a crowdsourced data collection in 53 fourth- and fifth-grade classrooms across 15 states in 2023–2025; 51 teachers appear in the transcribed corpus and 50 in the validation analyses below. Classrooms spanned all four Census regions and matched U.S. elementary schools in poverty, enrollment, and setting. Lessons were recorded through the year, transcribed automatically, and corrected by trained research assistants; on average 11.5 transcripts per teacher, with 137.8 teacher utterances each. Only lead-teacher utterances are scored, and one- and two-word utterances are excluded. The target practice is the *Ask* category of the Science Discourse Instrument (SDI; Fishman et al., 2017), adapted to the utterance level.

The validation subsample comprises 20 segments of roughly fifteen minutes from 12 teachers, each contributing one or two segments. Segments were coded with a SERA adaptation of the SDI, which characterizes science discussion through a set of discussion features and six discursive forms, three for the teacher and three for students; we use the teacher form ASK, open-ended questions about important scientific ideas whose answer is not already evident (typically *why* or *how* questions), as distinct from closed or factual questions with a single

expected answer. Three qualitative coders each read the segments and excerpted every teacher utterance they judged to contain an ASK move; the reference labels are the union of the coders' excerpts. Each excerpt was matched by its text to the utterance it was drawn from, and every teacher utterance inside a coded segment was labeled positive if a matched excerpt fell in it, yielding 787 hand-coded utterances, of which 152 (19.3%) are positives. Because teachers contributed unequal numbers of segments, all of a teacher's coded utterances are pooled into one set per teacher, so the validation design has no lesson replication; the lesson-level DCF components and the within-teacher prevalence variance therefore cannot be estimated here and are fixed by assumption when CRR is projected.

A Designed Experiment over Scoring Configurations

The goal is operational: to find an openly available LLM in the 7–12-billion-parameter range, small enough to run locally, together with a prompt whose measurement properties support applying one fixed configuration to the full corpus. We treated the pipeline's controllable stages as experimental factors and crossed them completely ([Table 1](#)). *Model*: eight instruction-tuned open-weight models, five from 2024 (Gemma-2-9B, Llama-3.1-8B, Mistral-7B, OLMo-2-7B, Qwen2.5-7B), one from 2025 (Granite-3.3-8B), and two from 2025–2026 (Gemma-4-12B, Qwen3-8B). *Instructions*: a two-line SHORT variant and a LONG variant with an expert-observer persona and rubric-style criteria. *Context*: the utterance alone (NOCTX), with the preceding teacher utterance (TLAG), or with the intervening student turns as well (TLAG+SLAG). *Examples*: none (NOEX), short synthetic positive exemplars (SYNTH), or one of three curated nine-example sets of real excerpts (SETA, SETB, SETC). The factorial yields $8 \times 2 \times 3 \times 5 = 240$ configurations, each applied to all 787 validation utterances with the single-pass restricted-label readout, so scoring is deterministic.

Applying the Criterion

For each configuration, each teacher's mean scores over gold-negative and gold-positive utterances give $\hat{A}_p$, $\hat{J}_p$, and $\hat{\pi}_p$; the across-teacher variances and covariances of these, disattenuated for within-teacher sampling error, estimate the teacher-level components, and their means estimate $\bar{A}$, $\bar{J}$, and μ_π . Two inputs cannot be estimated without lesson replication and are fixed in advance: the within-teacher prevalence variance is set equal to the between-teacher variance, consistent with the observation literature (Hill et al., 2012; Kane & Staiger, 2012), and lesson-level DCF is set to zero. The realization variance is computed from each configuration's pooled within-class variances at the planned lesson length. The planned design is four lessons of 150 utterances, and the floor is $\text{CRR} \geq .65$, approximately the reliability the Measures of Effective Teaching project achieved for human-scored observations with four observations per teacher (Kane & Staiger, 2012). The criterion thus asks the classifier to deliver, from four lessons, construct-relevant consistency comparable to the human-rated benchmark at the same budget. The floor is the user's choice in light of the intended use, not a property of the method: a low-stakes research use might accept less, and a consequential use of individual teacher scores should demand more, just as reliability standards for human-rated observation systems vary with the decisions they inform. With twelve teachers the ranking is a screening device, and near-ties should be treated as indistinguishable. The planned design also determines the outcome: no configuration clears the floor at two lessons and twenty do at eight, so the criterion ranks configurations for a specified deployment rather than certifying any absolutely.

Selected Configurations

Five of the 240 configurations meet the CRR_{zc} floor and enter the candidate set (Table 2; Figure 13). All five combine separation ($\bar{J} = .45$ to $.64$) with little detectable teacher-level DCF

($\lambda \geq .88$), the two properties CRR requires jointly. Configurations that excel on one property alone fail: the highest- λ configurations are low-separation classifiers whose clean decomposition protects almost no signal (an OLMo-2-7B configuration reaches $\lambda = 1.00$ with $\bar{J} = .12$ and CRR = .64), and several high-separation Gemma-4-12B configurations fall just short because teacher-level DCF consumes the error budget. No Qwen2.5-7B, Llama, Mistral, OLMo, or Granite configuration passes.

CRR_{zc} ranks the candidates; the panel decides which can be trusted, and two of the five survive. The leading Gemma-4-12B configuration fails the assumption its CRR rests on: $\delta_\beta = -.44$, the classifier's separation falls off for the teachers with the highest prevalence, and its covariance-inclusive CRR is .17. The TLAG+SLAG variant fails for the same reason, and the NOCTX/SYNTH variant, though within the δ_β tolerance, has a covariance-inclusive CRR of .56, below the floor. The survivors are the Qwen3-8B configuration ($\text{CRR}_{zc} = .68$; CRR = .68) and the Gemma-2-9B configuration ($\text{CRR}_{zc} = .66$; CRR = .80). The Qwen3-8B value carries a qualification: both of its teacher-level DCF components truncate to zero, so its CRR is an upper bound and its δ_β of zero means no covariance could be estimated. The Gemma-2-9B configuration is the better-behaved instrument by the panel's logic, with an estimable, positive δ_β . Both survivors have low intercepts ($\bar{A} = .13$ and .16), so the absolute-prevalence requirement does not prune further, whereas the rejected leader had the highest intercept of the five. We fixed the screens before inspecting which configurations they removed. They are use-determined choices. The δ_β tolerance of $\pm .20$ equals one sampling SD of the diagnostic at twelve teachers (.15 to .21 in the covariance simulation), so it rejects a configuration only when its estimated δ_β is farther from zero than sampling error alone would put it.

The full panel shows two patterns. First, the model families reach similar CRR_{zc} through different prompts (Gemma-2-9B with LONG instructions, lagged teacher context, and example set B; Gemma-4-12B with SHORT instructions, lagged teacher context, and example set A; Qwen3-8B with LONG instructions, lagged teacher context, and example set A), so prompt design must be selected jointly with the model rather than transferred. Second, saturation is common: 81% of configurations have negative δ_β , and the model that separates best on average, Gemma-4-12B (mean $CRR_{zc} = .49$ over its thirty prompts, against at most .32 for any other model), saturates in every one of its configurations (mean $\delta_\beta = -.30$). The theory predicts this pattern: CRR_{zc} is most optimistic for exactly the classifiers that separate best, which is why it ranks but does not certify.

Score Validation on the Application Corpus

The last step carries the two surviving configurations, Qwen3-8B (LONG/TLAG/SETA) and Gemma-2-9B (LONG/TLAG/SETB), into the application corpus: 141 transcripts covering 137 lessons by 50 teachers, with teachers contributing one to four lessons each (mean 2.7). This is the D study the criterion projected, and it serves two purposes. The first is to check the projections: whether the reliability realized on the corpus matches what the validation sample predicted. The second is a criterion-validity check against human observers who rated the same lessons in person: whether observers rate a lesson as higher in quality when the classifier, applied to the transcript of that lesson, finds a higher prevalence of open-ended questions.

At four lessons the realized coefficients are .87 for Qwen3-8B and .80 for Gemma-2-9B, against projections of .68 and .66 from the validation sample, and both projections fall within the bootstrap confidence intervals of the realized coefficients ($[.60, 1.00]$ and $[.49, .98]$). These are small-sample estimates, 50 teachers with an average of 2.7 lessons each, and the intervals are

correspondingly wide, so the application corpus is consistent with the projections rather than a sharp test of them. That the point estimates sit above the projections is expected: CRR projects construct-relevant reliability alone, whereas the realized coefficient is a replication reliability that also counts replication-stable classifier artifact as signal. Part of the gap also reflects the design assumption built into the projection: the projections split prevalence variance evenly between teachers and lessons, whereas the corpus splits it 51/49 for Gemma-2-9B but 63/37 for Qwen3-8B, the configuration whose realized coefficient most exceeds its projection. At the lesson budgets teachers actually contribute, reliability is lower (.78 and .67 at two lessons, .84 and .75 at three), so the operational lesson budget, not the classifier, is now the binding constraint.

The scores also agree with external information in the expected direction. The two configurations agree on teacher standing ($r = .93$; Spearman $\rho = .90$), and soft and thresholded scores are nearly interchangeable at the teacher level ($r \geq .99$), as the model predicts when classes are well separated. Trained observers rated a subset of the recorded lessons in person on the SERA direct observation rubric, whose high-inference indicators rate broader constructs on five-point scales. Teacher-level prevalence of open-ended questioning correlates most strongly with *Teaching for Scientific Understanding* ($r = .49$ and $.46$ for Qwen3-8B and Gemma-2-9B), next with *Overall Discourse Opportunities*, the rubric construct whose definition includes the scored practice but which rates student participation in discussion as well as the teacher's questioning ($r = .33$ and $.31$), then with whole-class *Student Interest* ($r = .33$ and $.29$), and least with the engagement of a single target student ($r = .20$ and $.11$). The gradient is what a convergent check should show: the association is strongest for the rating that describes the teacher's own instructional moves, weaker for ratings that depend partly on how students participate and engage, and weakest for a rating tied to one student's behavior.

Two cautions bound what the application corpus shows. These are replication reliabilities; they certify consistency across lesson samples, not the absence of stable classifier artifact, and the DCF diagnostics that CRR incorporates remain estimable only on hand-coded data. And the corpus's teachers contribute few and unequal lessons, so the variance components are imprecisely estimated. Lesson-level DCF diagnostics on a multi-lesson coded sample are the necessary next step before operational use.

Discussion

Utterance-level agreement establishes that an LLM classifier can label teaching moves; it does not establish that the teacher scores built from those labels are reliable. In traditional observation systems these two questions are linked by design: scores are inconsistent from lesson to lesson and from rater to rater, so dependable teacher scores are built by sampling lessons and raters and averaging their disagreements away. An LLM classifier changes the structure of the problem. Scores are still inconsistent over lessons, but the classifier is a single fixed rater that returns the same labels every time it reads a transcript, so rater-to-rater inconsistency disappears. Whatever errors the classifier makes are frozen into every replication of the procedure, so an accurate-looking classifier can still produce unreliable teacher scores, and the methodological question shifts from sampling enough rater perspectives to selecting the one classifier whose fixed errors are benign.

Two sources of classifier error therefore need to be considered when selecting the classifier that will fix a scoring procedure, and they behave differently. Random classifier error, the utterance-level noise in the classifier's scores, behaves like ordinary lesson-to-lesson noise: like the disagreements among human raters, it averages out, though through the sampling of utterances and lessons rather than the sampling of raters, and it can be reduced with more and

longer lessons. Systematic error, differential classifier functioning across teachers, cannot be averaged out by any amount of sampling, because it is reproduced identically in every replication. It enters the universe score, where it inflates the generalizability coefficient while construct-relevant reliability collapses; more lessons make the certification more confident and no more valid.

To certify scores, and to guide the selection of a classifier, we proposed projected construct-relevant reliability. CRR combines classifier accuracy, differential classifier functioning, and the sampling design into a single coefficient that answers the deployment question directly: under a planned design, what share of the variance in teachers' scores will reflect true differences in their practice rather than differences in how the classifier treats them? Formally, CRR is a D-study coefficient in the generalizability-theory sense, a projection of reliability under a planned number and length of lessons, but one that carries the classifier's fixed errors into the coefficient rather than treating them as sampled error that a larger design would dilute.

The recommendation, then, mirrors long-standing practice for human-scored observation systems: before deploying a new measure, run a generalizability study to learn where the error lives and a decision study to design against it (Brennan, 2001; Cronbach et al., 1972; Hill et al., 2012). The case is stronger here. LLM performance varies widely across models, prompts, and examples, and small configuration changes moved projected reliability across its full range in our experiment, so intuition about which configuration will score reliably is a poor guide. A G-study of the kind we describe turns that variability into a measurement-design decision made on evidence. And it enforces the distinction that organizes this paper: accuracy is a property of

utterance-level labels, reliability is a property of object-level scores, and only an analysis conducted at the level of the score can certify the score.

Certifying scores at the object level yields a workflow. A researcher hand-codes a small validation sample, runs a designed experiment over candidate models and prompts, estimates the variance components that CRR requires, and projects reliability under the design they actually intend to field. Selection then happens on the reliability of the object-level score rather than on utterance-level accuracy alone, the companion panel flags configurations whose ranking rests on a violated assumption, and a downstream check on independent data confirms that the projection held. None of these steps requires more than the validation data a careful accuracy evaluation would collect anyway; what changes is the level at which the evidence is read.

Limitations

First, the reliability formulas are derived for scores built from binary classifications applied to utterances. Many observation constructs are scored on ordinal scales, and classifiers can emit ordinal (low, medium, high) or nominal labels; we leave the extension to these cases to future work. Second, we analyzed a single teaching move. In practice, several teaching moves may be classified separately and then aggregated into a holistic score. The measurement model and the mvgstudy machinery are multivariate, but we did not examine how differential classifier functioning covaries across moves or how it propagates into a composite. Future work is needed on the reliability of composite scores. Third, the validation sample was small. With twelve teachers, CRR was estimated imprecisely, and the sample lacked lesson replication, so the lesson-level components entered as design assumptions rather than estimates; the human codes were themselves treated as truth. Larger validation samples with replicated lessons are needed to

establish the sampling behavior of CRR and, in particular, whether projected CRR predicts realized CRR well enough to bear the selection weight we place on it.

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Appendix: Sources of Reported Results

Every table, figure, and number in the paper traces to a script and an output file in the project repository ([Table A1](#)). Simulation scripts live in `simulation/code/` with outputs in `simulation/output/` (figures in `simulation/output/figures/`); scoring and selection scripts live in `sdi_scoring/`. Derivations are in the technical appendix.

Table A1. Sources of the results reported in the paper.

Result	Reported in	Script(s)	Output file(s)
Pipeline schematic	Figure 1	—	documents/presentations/code/figures/pipeline.png
Hand-coded validation subsample: 20 segments, 12 teachers, 787 utterances, 152 positives	Data	sdi_scoring/production_pipeline_v3/pipeline/ (gold-label stage)	production_pipeline_v3/outputs/gold_labels/match_audit.json
Simulation reference values and calibration	Simulation Studies	00_obtain_reference_values.do	reference_values.csv
Constant-functioning simulation	Simulation Studies; Figures 2–5	01_simulation_1.do, 03_simulation_1_analyses.do	simulation1_cells.csv; fig1_reliability_by_J_length.pdf, fig2_intercept_by_nL.pdf, fig5_oneway_sweeps.pdf, fig3_soft_vs_hard.pdf
DCF simulation	Simulation Studies; Figures 6–9	02_simulation_2.do, 04_simulation_2_analyses.do	simulation2_cells.csv; s2fig1_person_dcf_by_tau.pdf, s2fig2_locus_by_nL.pdf, s2fig3_theory_overlay.pdf, s2fig4_reordering_lambda.pdf
Within-lesson dependence simulation	Simulation Studies (one paragraph; technical appendix)	05_simulation_3_dirichlet.do, 06_simulation_3_analyses.do	simulation3_cells.csv; s3fig1_dirichlet_naive_gap.pdf
Covariance simulation	Simulation Studies; Figures 10–12	08_simulation_4.do, 09_simulation_4_analyses.do	simulation4_cells.csv, simulation4_summary.csv; s4fig1_divergence.pdf, s4fig2_lesson.pdf,

Result	Reported in	Script(s)	Output file(s)
Configuration experiment: scoring of the validation subsample under 240 configurations	Designed Experiment; Table 1	sdi_scoring/production_pipeline_v3/pipeline/ (factor levels in config.py)	s4fig3_estimation.pdf production_pipeline_v3/outputs/training/train_predictions_long.csv
Per-configuration panel, candidate set, survivors	Applying the Criterion; Selected Configurations; Table 2 ; Figure 13	sdi_scoring/application/code/00_config_ranking_panel.do (mvgstudy/mvcrr v1.5.1)	sdi_scoring/application/validation_output/crr_panel_q00_ASK.csv, crr_panel_selection_q00_ASK.csv, crr_panel_ranking_q00_ASK.pdf
Application-corpus scoring of the two selected configurations	Score Validation	sdi_scoring/application/code/01_prepare_data.py, 02_score_application.py, 03_combine_scores.py, 04_collapse_to_lessons.do	lesson-level score files under sdi_scoring/application/
Teacher-level validation: variance components, realized reliability and bootstrap CIs, correlations with observer ratings	Score Validation	sdi_scoring/application/code/05_validate_teacher_scores_v2.do	validation_output/teacher_scores_selected_ASK.csv

Tables

Table 1. Design factors of the configuration experiment (complete factorial, 240 configurations).

Factor	Levels	Description
Model	8	Gemma-2-9B, Llama-3.1-8B, Mistral-7B, OLMo-2-7B, Qwen2.5-7B, Granite-3.3-8B, Gemma-4-12B, Qwen3-8B (instruction-tuned, open-weight, run locally)
Instructions	2	SHORT (two-line task statement); LONG (expert-observer persona with rubric-style criteria)
Context	3	NOCTX (utterance only); TLAG (+ preceding teacher utterance); TLAG+SLAG (+ intervening student turns)
Examples	5	NOEX (none); SYNTH (synthetic positive exemplars); SETA, SETB, SETC (nine real excerpts each, 6 positive / 3 negative)

Table 2. The five configurations with $CRR_{zc} \geq .65$ at four lessons of 150 utterances, with the companion panel. Screens: $\bar{J} \geq .30$, $|\delta_\beta| \leq .20$, $CRR \geq .65$.

Configuration	CRR_{zc}	δ_β	CRR	$\bar{J}$	$\bar{A}$	Outcome
Gemma-4-12B, SHORT / TLAG / SETA	.70	-.44	.17	.62	.31	fails δ_β screen
Qwen3-8B, LONG / TLAG / SETA	.68	.00 (upper	.68	.45	.13	selected

Configuration	CRR_{zc}	δ_β	CRR	$\bar{J}$	$\bar{A}$	Outcome
		bound)				
Gemma-2-9B, LONG / TLAG / SETB	.66	+.15	.80	.53	.16	selected
Gemma-4-12B, SHORT / TLAG+SLAG / SETA	.66	-.39	.26	.64	.24	fails δ_β screen
Gemma-4-12B, LONG / NOCTX / SYNTH	.66	-.13	.56	.58	.11	fails CRR screen

Figures

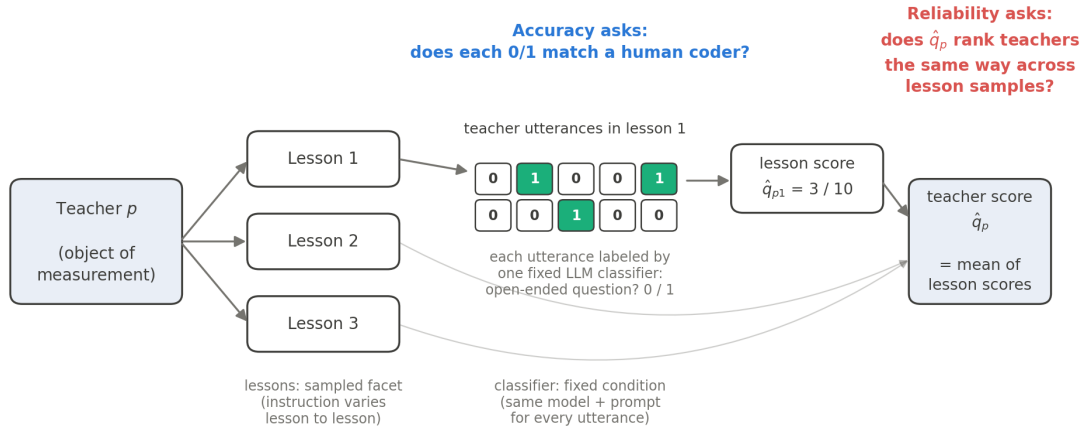


Figure 1. From utterance labels to a teacher score. Each lesson's teacher utterances are labeled by one fixed LLM classifier; the lesson score is the mean of the utterance scores and the teacher score is the mean of the lesson scores. Lessons are a sampled facet, the classifier a fixed condition. Accuracy is a question about utterance labels; reliability is a question about teacher scores.

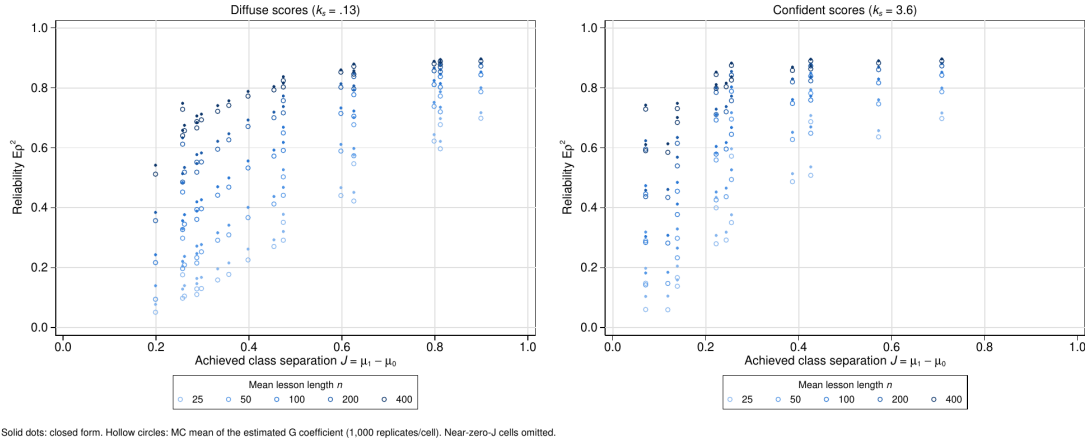


Figure 2. Reliability by achieved class separation J and mean lesson length under constant classifier functioning; solid dots show the closed form and hollow circles the Monte Carlo mean of the estimated coefficient. Separation matters most at short lessons.

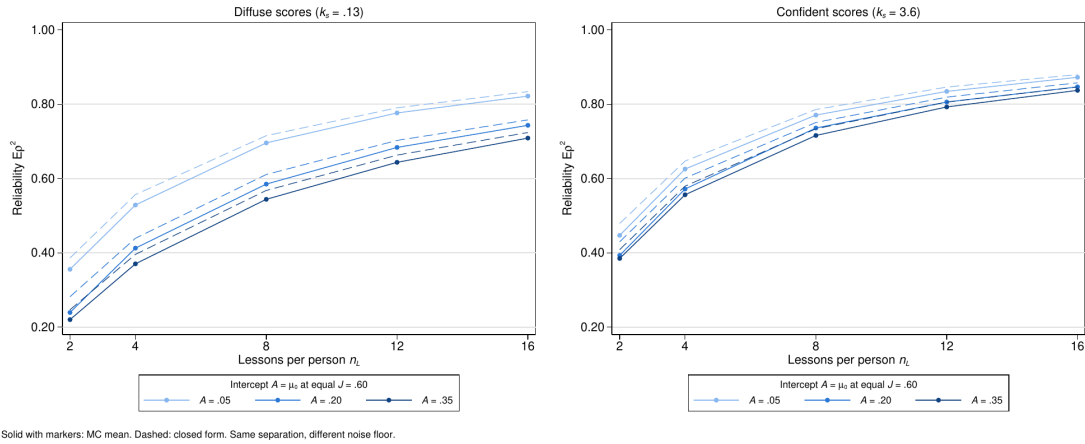


Figure 3. Reliability by lessons per teacher at three negative-class intercepts ($A = .05, .20, .35$) with class separation fixed at $J = .60$: equal separation, different noise floor.

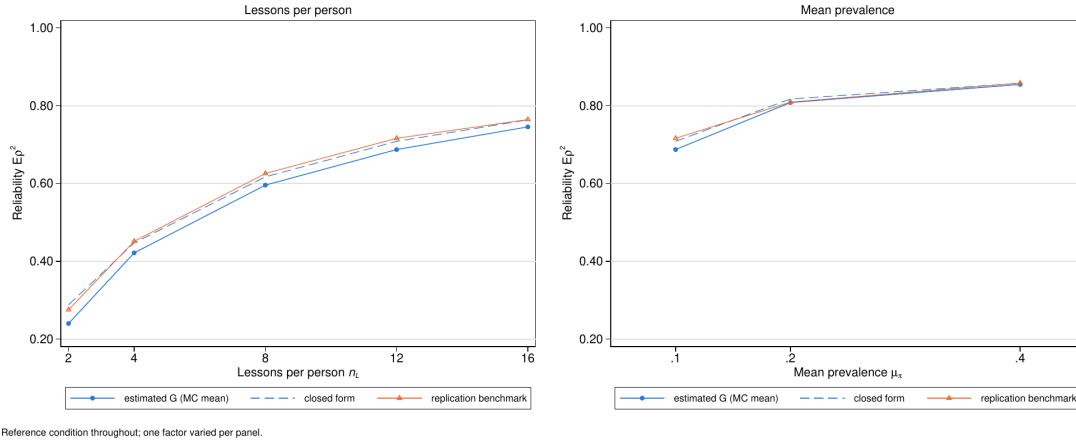


Figure 4. One-way sweeps around the reference condition: reliability by lessons per teacher (left) and mean prevalence (right), comparing the estimated coefficient, the closed form, and the replication benchmark.

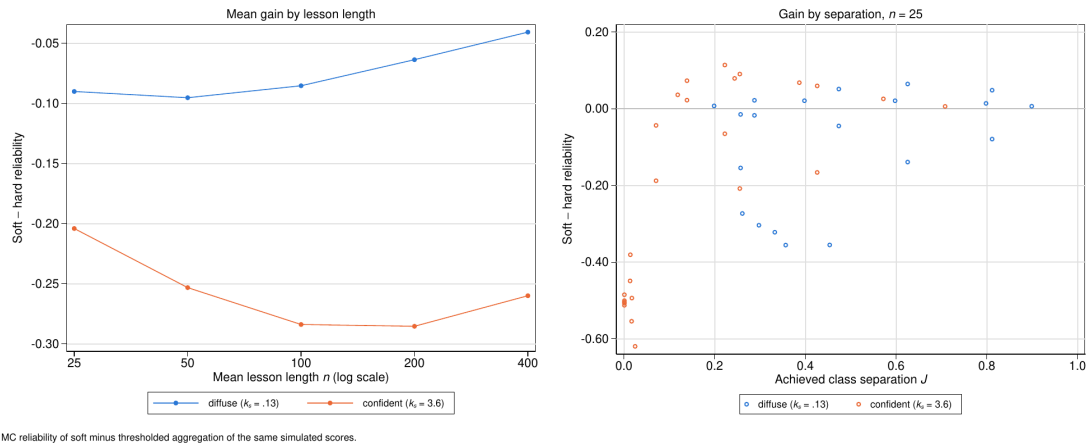


Figure 5. Soft minus thresholded (hard) reliability for the same simulated scores at matched sensitivity–specificity operating points, by lesson length (left) and by achieved separation at 25 utterances per lesson (right).

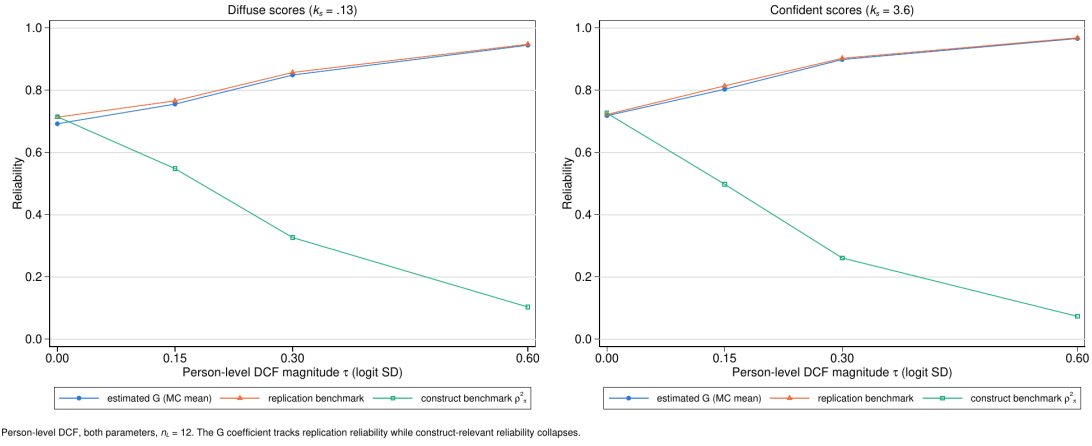


Figure 6. Teacher-level DCF: the estimated coefficient tracks the replication benchmark while construct-relevant reliability collapses as DCF grows.

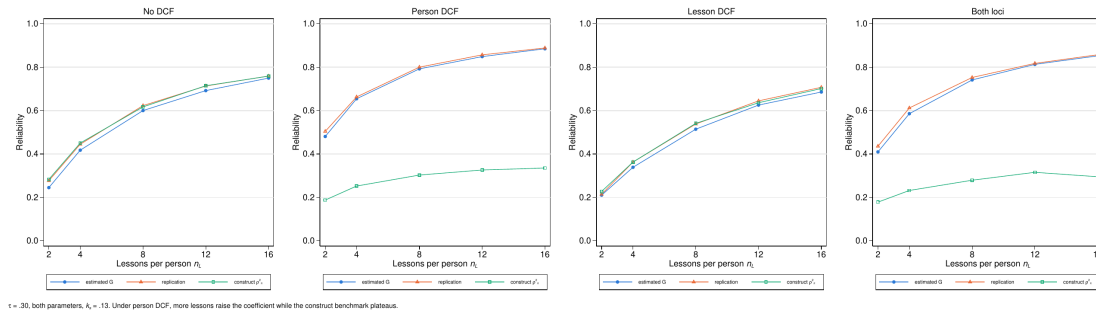


Figure 7. Reliability by lessons per teacher under no DCF, teacher-level DCF, lesson-level DCF, and both; more lessons widen the gap under teacher-level DCF.

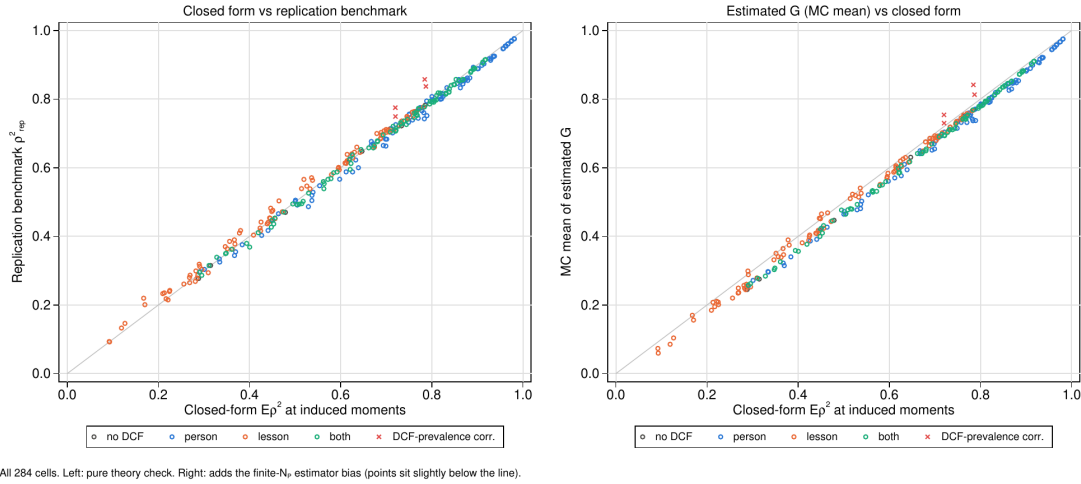


Figure 8. The DCF-extended closed forms overlaid on the simulation benchmarks across the DCF simulation's cells; deviations concentrate where the zero-covariance assumption is violated.

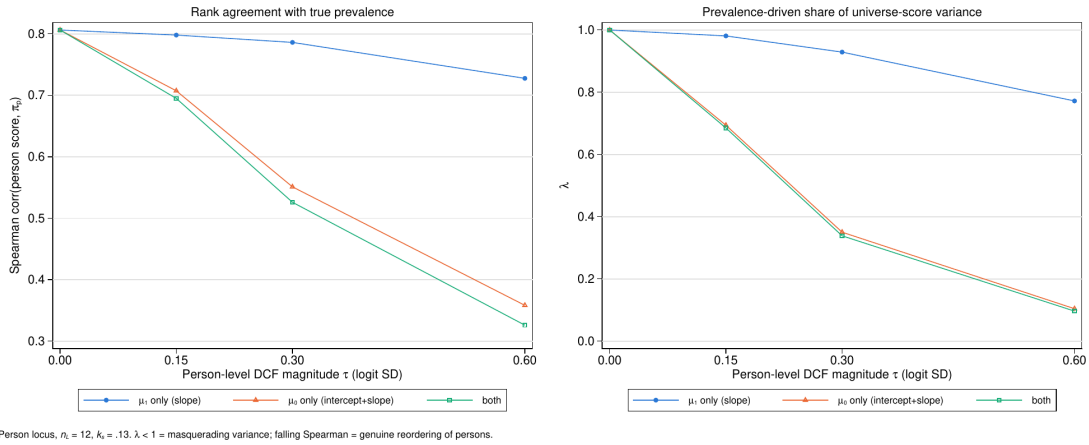


Figure 9. Which parameter drifts matters: rank agreement with true prevalence (left) and the prevalence-driven variance share λ (right) by DCF magnitude, with variation entering through the positive-class mean only, the negative-class mean only, or both (teacher level, twelve lessons).

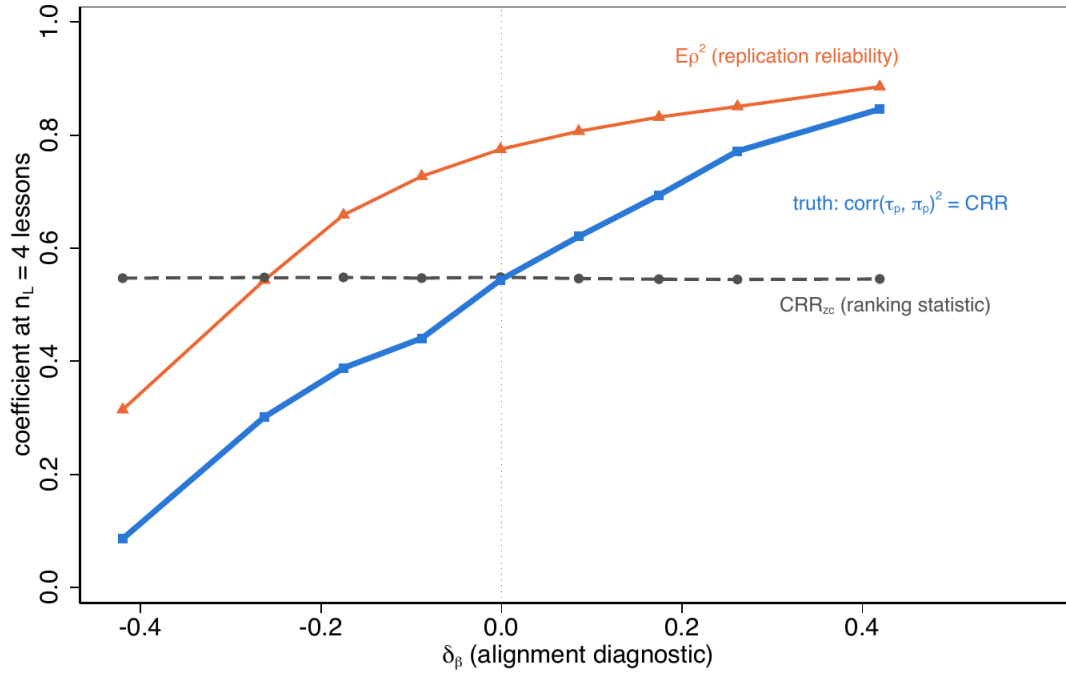


Figure 10. Covariance between classifier functioning and prevalence: the realized squared correlation with true prevalence (truth), the replication reliability, and CRR_{zc} along the alignment diagnostic δ_β .

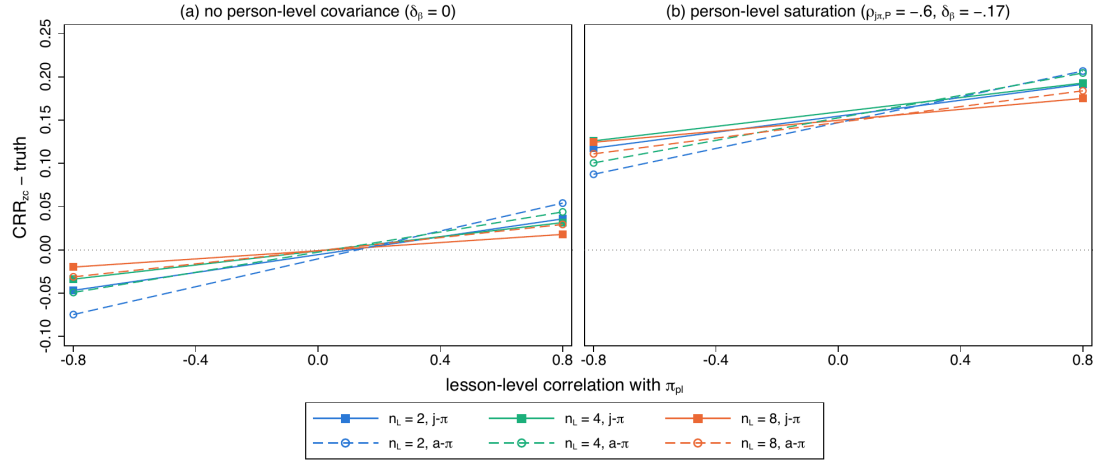


Figure 11. Overstatement of the truth by CRR_{ZC} when covariance is present only at the lesson level (a; $\delta_\beta = 0$ in every cell) and when teacher-level saturation is also present (b), by lessons per teacher; solid: separation–prevalence route, dashed: intercept–prevalence route.

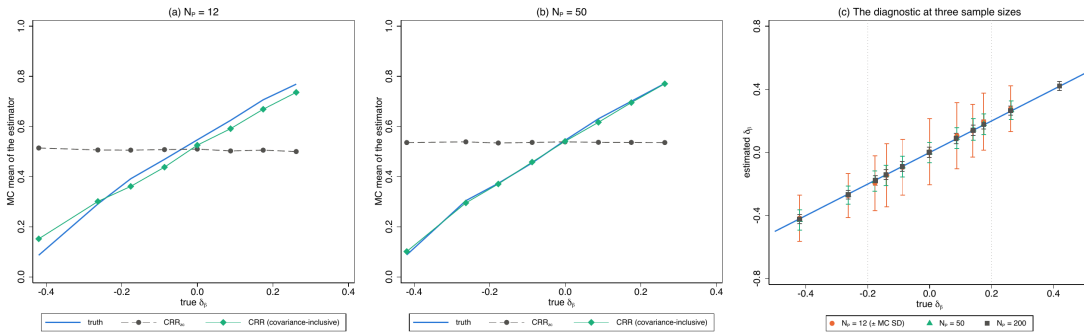


Figure 12. Monte Carlo means of CRR_{ZC} and CRR against the truth along δ_β at 12 (a) and 50 (b) teachers, and the estimated alignment diagnostic ($\pm$ one Monte Carlo SD) against its true value at three sample sizes (c).

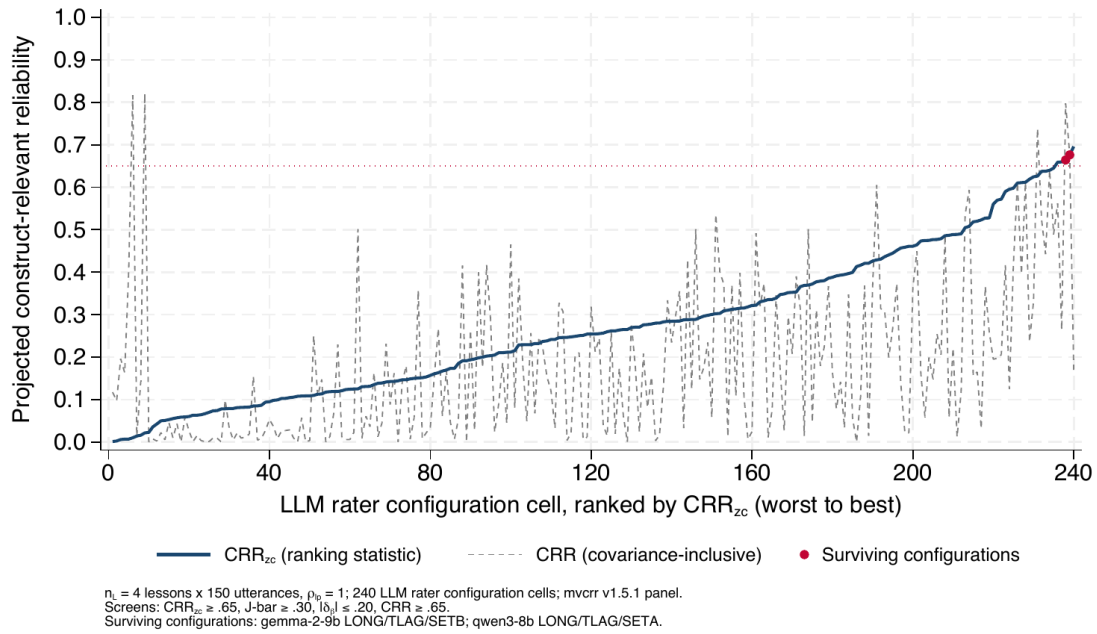


Figure 13. The 240 configurations ranked by CRR_{zc} at the planned design, with CRR as a thin dashed line and the two selected configurations marked.